

$$a + jb \cdot \left(\frac{V_o}{V_i} \right) = \left(\frac{1}{\sqrt{2}} \right)$$

$$\left(\frac{1}{\sqrt{2}} \right)^2 = \left(\frac{1}{\sqrt{1 + \omega^2 R^2 C^2}} \right)^2$$

$$\frac{1}{2} = \frac{1}{1 + \omega^2 R^2 C^2}$$

$$2 = 1 + \omega^2 R^2 C^2$$

$$\sqrt{1} = \sqrt{\omega^2 R^2 C^2}$$

$$1 = \omega R C$$

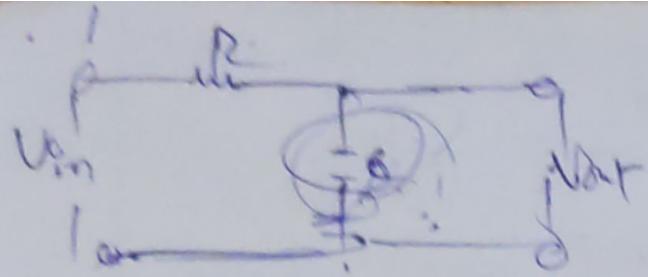
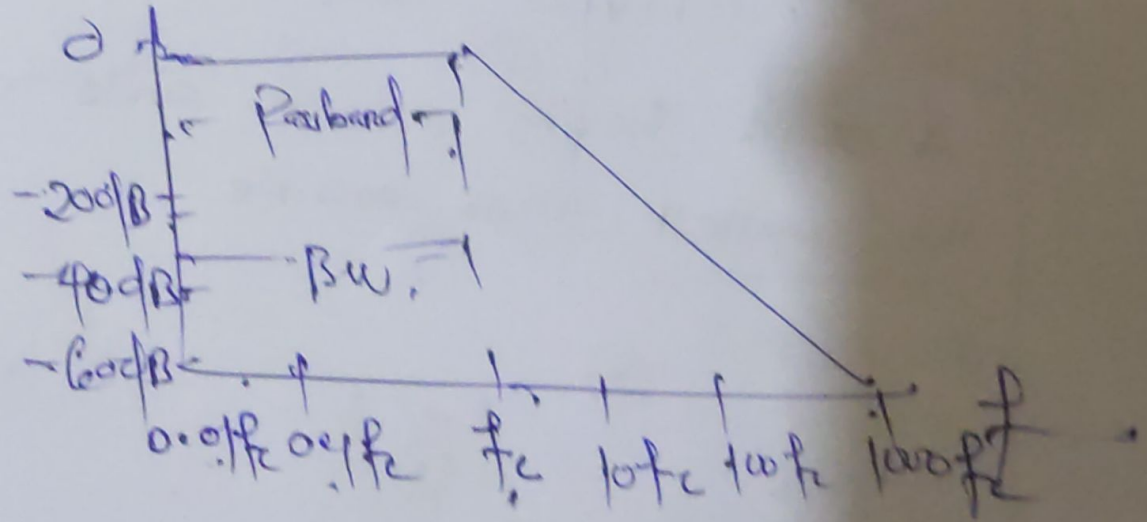
$$\omega = \frac{1}{RC}$$

$$2\pi f = \frac{1}{RC}$$

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$$

Frequency filter response
Gain



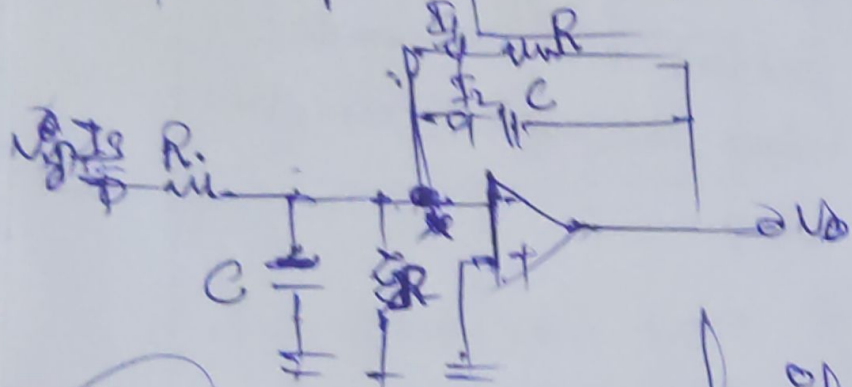
$$V = I \left(\frac{X_c}{R + X_c} \right)$$

$$R = X_c$$

$$R = \frac{1}{2\pi f C}$$

$$f_c = \frac{1}{2\pi RC}$$

Active low pass filter design



Derivation

At node X,

$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_o}{R} + \frac{V_o}{X_c} + \frac{V_i}{R} = 0$$

$$V_o \left(\frac{1}{R} + \frac{1}{X_c} \right) = -\frac{V_i}{R}$$

$$V_o \left(1 + \frac{R}{X_c} \right) = -V_i$$

$$X_c = \frac{1}{j\omega C}$$

$$V_o (1 + j\omega R C) = -V_i$$

$$\frac{V_o}{V_i} = \frac{-1}{1 + j\omega R C}$$

$$\left| \frac{V_o}{V_i} \right| = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$$

Characteristics of an ideal filter:

1. Lowpass filters pass low frequencies and attenuate high frequencies.
2. Highpass filters pass high frequencies and attenuate low frequencies.
3. Bandpass filters pass certain band of frequencies.
4. Bandstop filters attenuate a certain band of frequencies.

Basic filter response

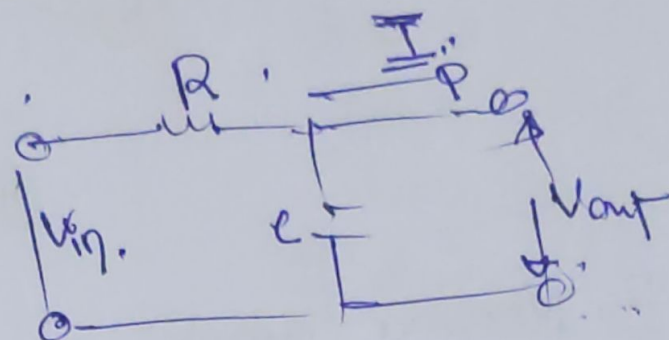
$$V_c = I \left(\frac{X_c}{R + X_c} \right)$$

$$X_c = R_c$$

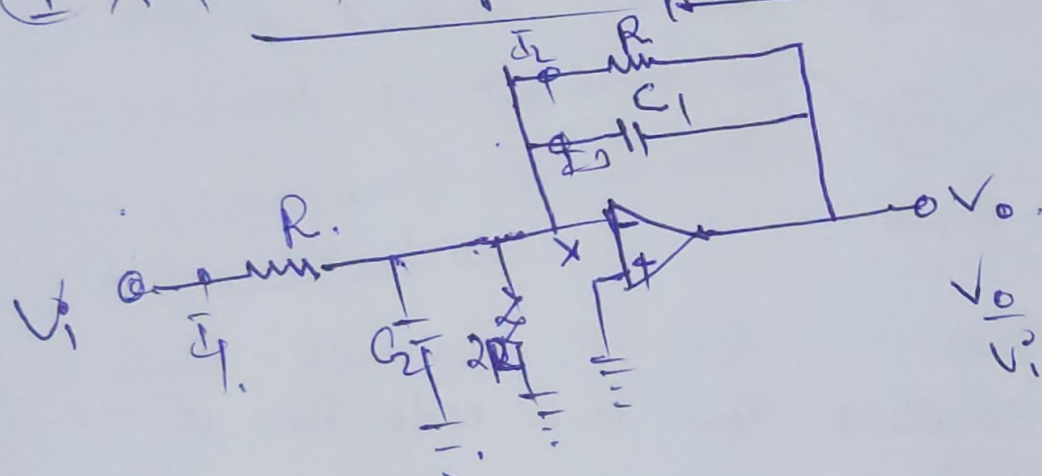
$$R_c = \frac{1}{2\pi f_c C}$$

$$f_c = \frac{1}{2\pi R C}$$

Better filter must have small bandwidth.



① Active low pass filter design



$$\frac{V_o}{V_i} = \frac{-X_c}{X_c + R}$$

$$\text{but } X_c = \frac{1}{j\omega C}$$

$$\frac{V_o}{V_i} = - \frac{1/j\omega C}{1/j\omega C + R}$$

$$\frac{V_o}{V_i} = \frac{-1}{1 + j\omega RC}$$

$$\left| \frac{V_o}{V_i} \right| = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$$

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$$

$$\omega^2 R^2 C^2 + 1 = 2$$

KCL at X

$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_o}{R} + \frac{V_o}{R} + \frac{V_o}{X_c} = 0$$

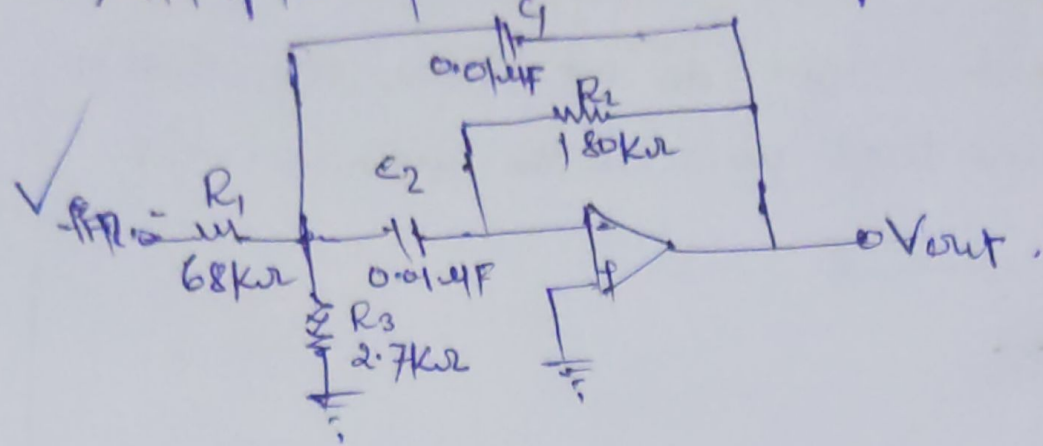
$$\frac{V_o}{R} = -V_o \left[\frac{1}{R} + \frac{1}{X_c} \right]$$

$$X_c = \frac{1}{j\omega C}$$

$$\frac{V_o}{R} = -V_o \left[\frac{X_c + R}{R X_c} \right]$$

$$V_i = -V_o \left[\frac{X_c + R}{X_c} \right]$$

Multiple-feedback Bandpass filter.



$$f_0 = \frac{1}{2\pi \sqrt{(R_1 R_3) R_2 C_1 C_2}}$$

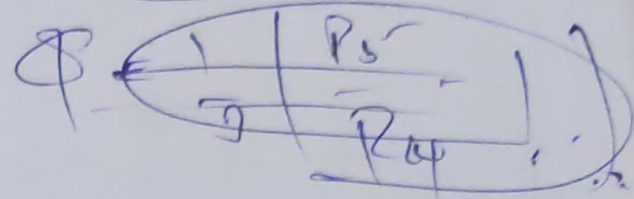
$$A_0 = R_2 / 2R_1$$

$$Q = \pi f_0 C R_2$$

$$BW = f_0 / Q$$

Application of filter.

1. Noise rejection.
2. Interference Rejection.
3. Anti-Alias filtering



Advantages.

- Accurate signal control.
- Enhanced performance.
- Enhancement of signal.
- Personalization.
- Broad Range of uses

Disadvantages.

- Component tolerance.
- Active filters nonlinearities.
- loss of signal.
- amount of power used in active filters.
- Complicated circuitry and careful design are needed for advanced filters.

Applications

1. Digital electronics.
2. Medical electronics.
3. Instrumentation and measurement.
4. Power supplies
5. Audio processing.
6. Communication system.
7. Signal processing.

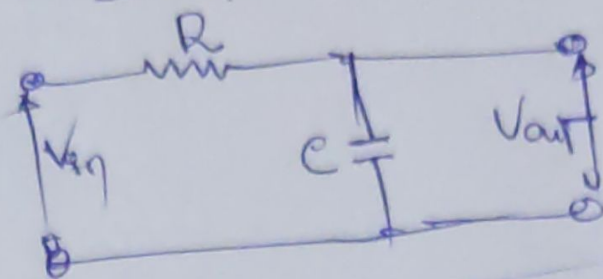
$$Q = \frac{1}{DF}$$

Filters

723

- Basic filter response
- Characteristics
- Low pass
- High pass
- Band pass
- Band stop
- Applications

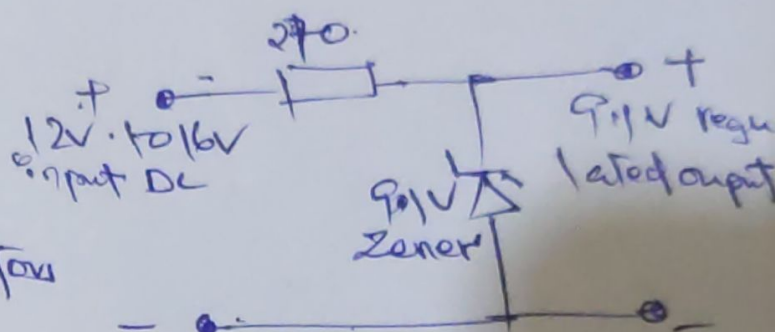
Circuit diagram



pole

Regulators

- Describe voltage regulation
- Line regulation and load regulation
- Basic linear shunt & linear series regulators
- Basic switching regulator
- IC voltage regulators



FILTERS

Filter is a circuit that passes certain frequencies and rejects all other frequencies.

Types of filters

- Passband. The range of frequencies that pass through the filter.
- Stopband. The range of frequencies that the filter attenuates.
- Cut-off frequencies.
- Bandwidth.
- Phase transition.
- Attenuation slopes.

- (a) Passive filters - Resistors,
Capacitors,
Inductors

- (b) Active filters - Opamps, in addition of resistor & capacitor.

$$\omega^2 R^2 C^2 = 1$$

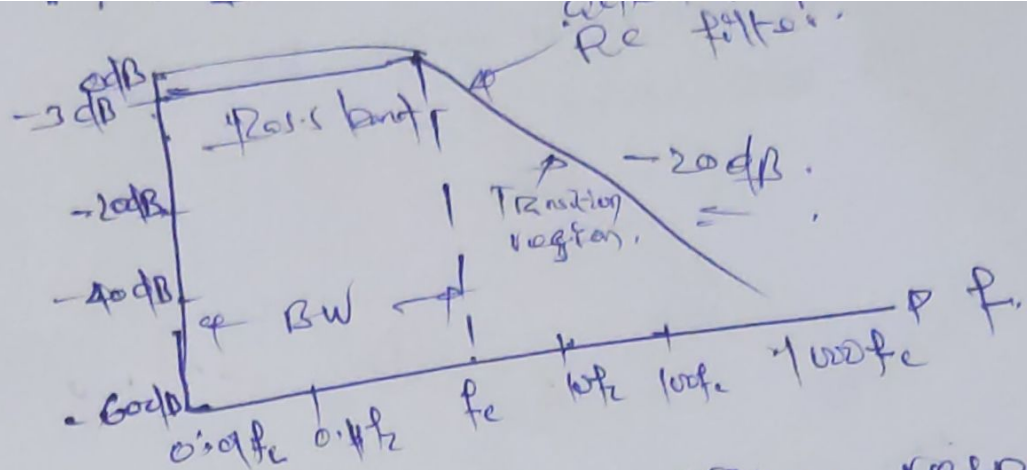
$$\omega^2 = \frac{1}{R^2 C^2}$$

$$\omega_c = \frac{1}{RC}$$

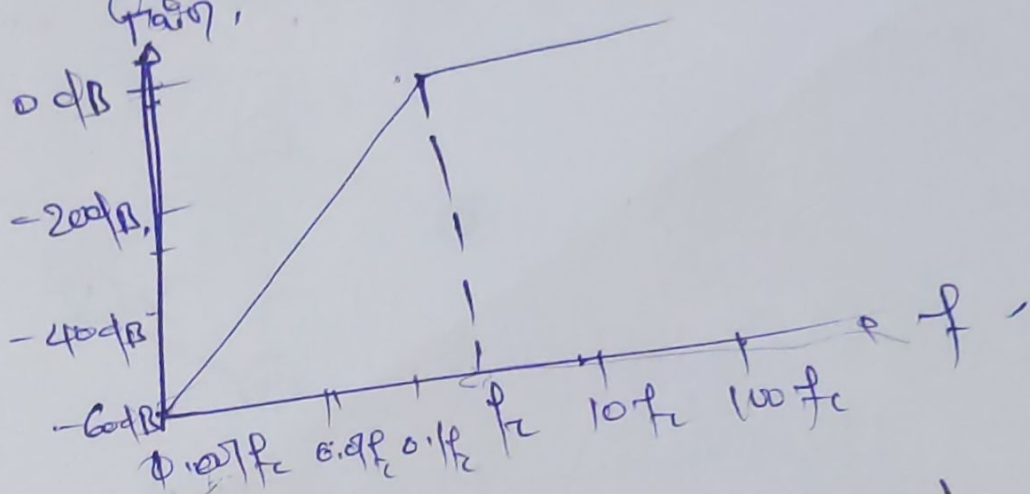
$$2\pi f_c = \frac{1}{RC}$$

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$$



High frequency filter response



Bandpass filter response

$$BW = f_{c2} - f_{c1}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$Q = \frac{f_0}{BW}$$

Quality factor

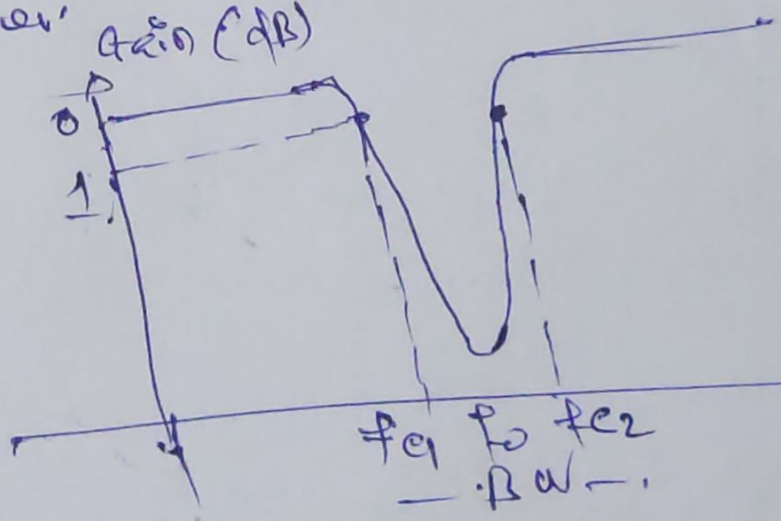
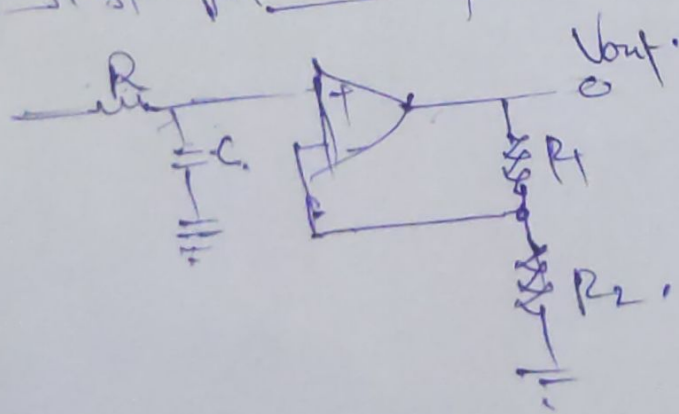
Narrow band filter

Eg. $Q > 10$ - Narrow band filter

$Q < 10$ - Wide band filter

Bandstop filter response

Simple active low pass circuit



$$Q = \frac{1}{DF}$$

damping factor

$$DF = 2 - \frac{R_1}{R_2}$$

$$\nabla \cdot \vec{v} \times \vec{A} = 0$$

$$\vec{v} = \frac{\partial}{\partial x} q_x + \frac{\partial}{\partial y} q_y + \frac{\partial}{\partial z} q_z$$

$$\vec{v} = \frac{\partial}{\partial x} q_x + \frac{\partial}{\partial y} q_y + \frac{\partial}{\partial z} q_z$$

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$$

$$\vec{v} \cdot \vec{v} = \left(\frac{\partial}{\partial x}\right)^2 + \left(\frac{\partial}{\partial y}\right)^2 + \left(\frac{\partial}{\partial z}\right)^2$$

Flux

$$A = f \cos \phi \hat{a}_y + z \sin \phi \hat{a}_z$$

$$\vec{v} = \frac{\partial}{\partial \phi} \hat{a}_\phi + \frac{1}{f} \frac{\partial}{\partial \phi} q_\phi + \frac{\partial}{\partial z} q_z$$

$$\vec{v} \cdot \vec{A} = \frac{\partial}{\partial \phi} (f \cos \phi) + 0 + \frac{\partial}{\partial z} (z \sin \phi)$$

$$= \frac{\partial}{\partial \phi} \cos \phi$$

$$\int \vec{v} \cdot \vec{A} = \int \cos \phi d\phi$$

$$= \cos \phi \Big|_0^2$$

$$= 2 \cos \phi$$

$$= 2(\cos 60^\circ - \cos 0^\circ)$$

$$= 2(-0.5)$$

$$= -1$$

Question

$$f_c = 15 \text{ kHz} = f_0$$

$$BW = 1 \text{ kHz}$$

$$Q = \frac{f_0}{BW} = \frac{15 \text{ kHz}}{1 \text{ kHz}}$$

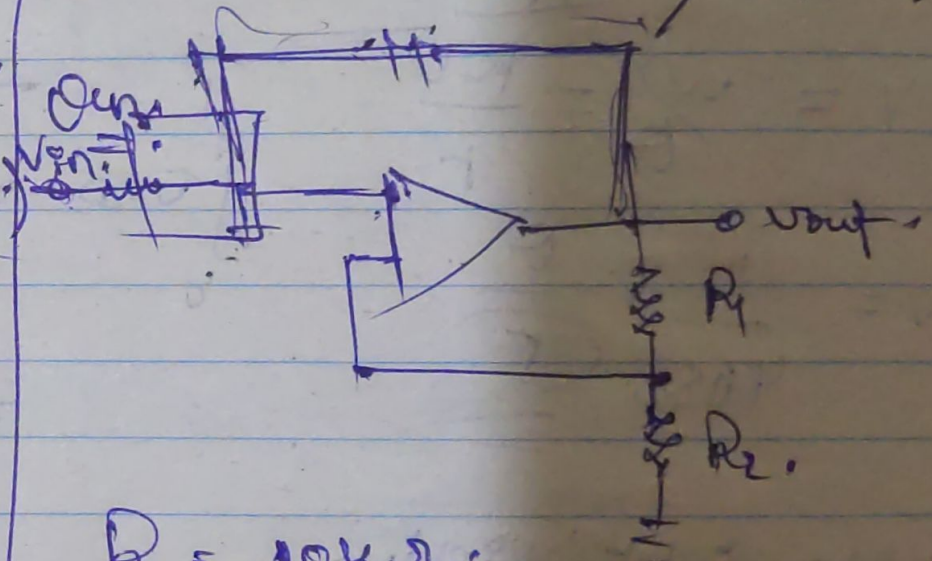
$$Q = 15$$

$Q > 10$ — narrow band

$Q < 10$ — wide band

$$Q, DF = \frac{R_1}{R_2}$$

$$A_{v(EMP)} = \frac{1}{B} = \frac{1}{R_2 / (R_1 + R_2)}$$



$$R_2 = 10 \text{ k}\Omega$$

$$R_1 = \text{required}$$

AT maximally flat Butterworth response.

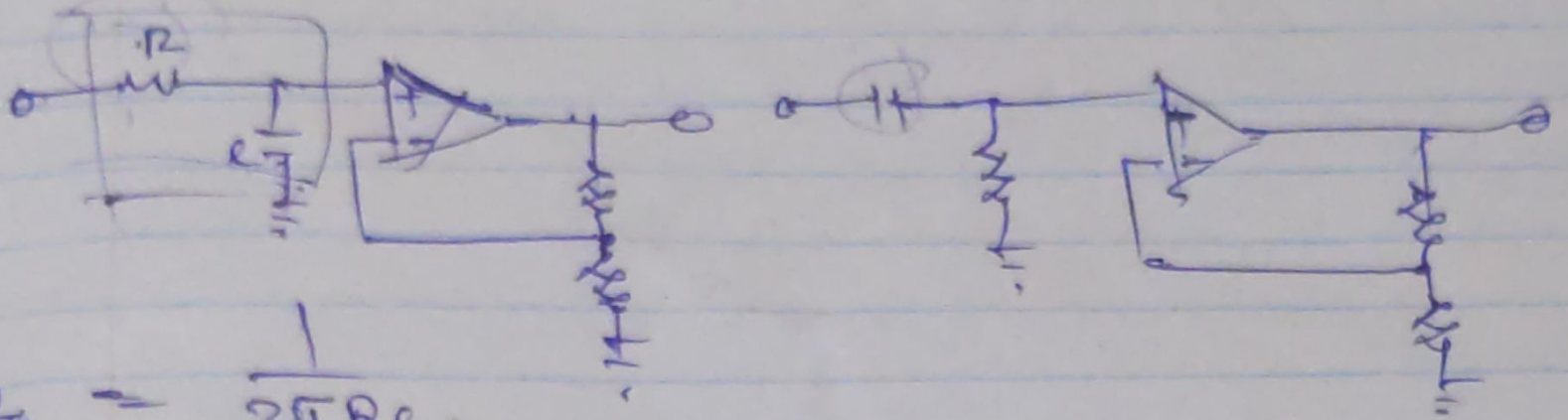
$$\frac{R_1}{R_2} = 0.586$$

$$R_1 = 0.586 R_2 = 0.586 \times 10 \text{ k}\Omega$$

$$R_1 = 5.86 \text{ k}\Omega$$

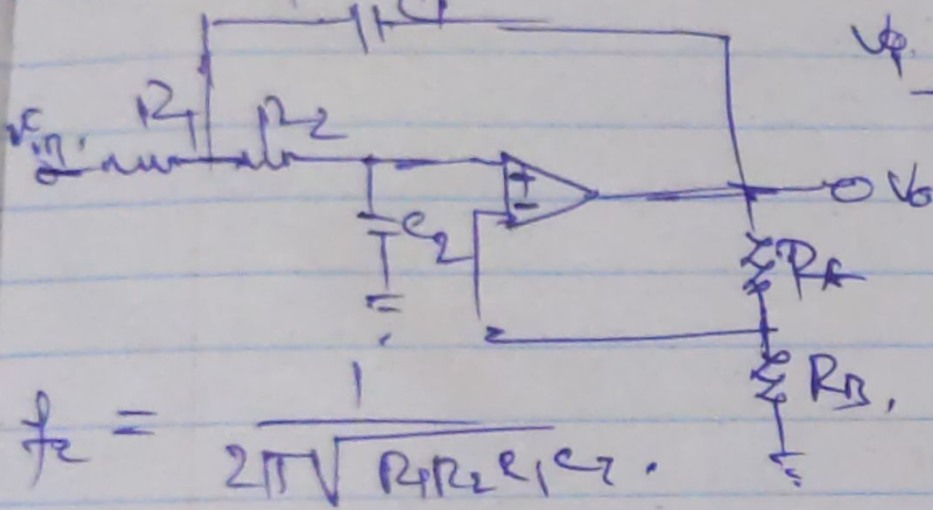
$$DF = Q = \frac{R_1}{R_2}$$

single pole filter, low filter, high filter



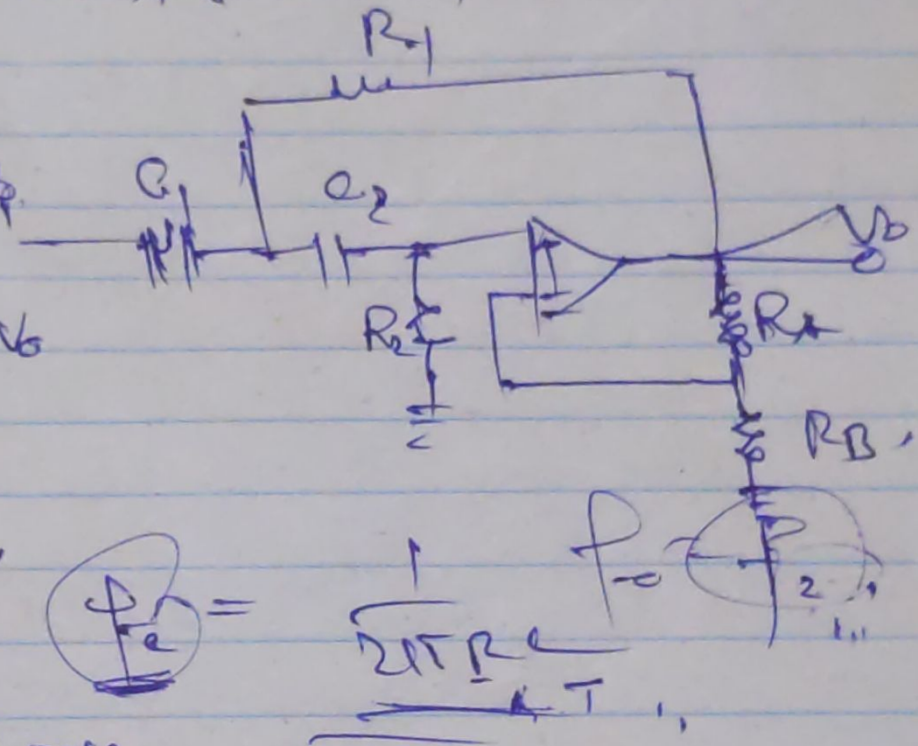
$$f_c = \frac{1}{2\pi RC}$$

Double pole low filter



$$f_c = \frac{1}{2\pi \sqrt{R_1 R_2} C_1 C_2}$$

high filter



$$f_c = \frac{1}{2\pi RC}$$

Multiple-feedback band pass

$$f_c = \frac{1}{2\pi \sqrt{(R_1/R_2) R_3} C_1 C_2}$$

$$f_c = \frac{1}{2\pi C \sqrt{(R_1/R_2) R_3}}$$

$$(R_1/R_2) R_3 = \left(\frac{R_1 R_2}{R_1 + R_2} \right) R_3$$

$$A_0 = \frac{R_3}{2R_1}$$

$$BW = \frac{f_0}{Q}$$

$$Q = \frac{f_0}{BW}$$

$$BW = \frac{f_0}{Q}$$

$$Q = \frac{1}{3} \left(\frac{R_5}{R_6} + 1 \right)$$

passband

$$Q = \pi f_0 C R_2 - \text{multiple feedback}$$

bandwidth